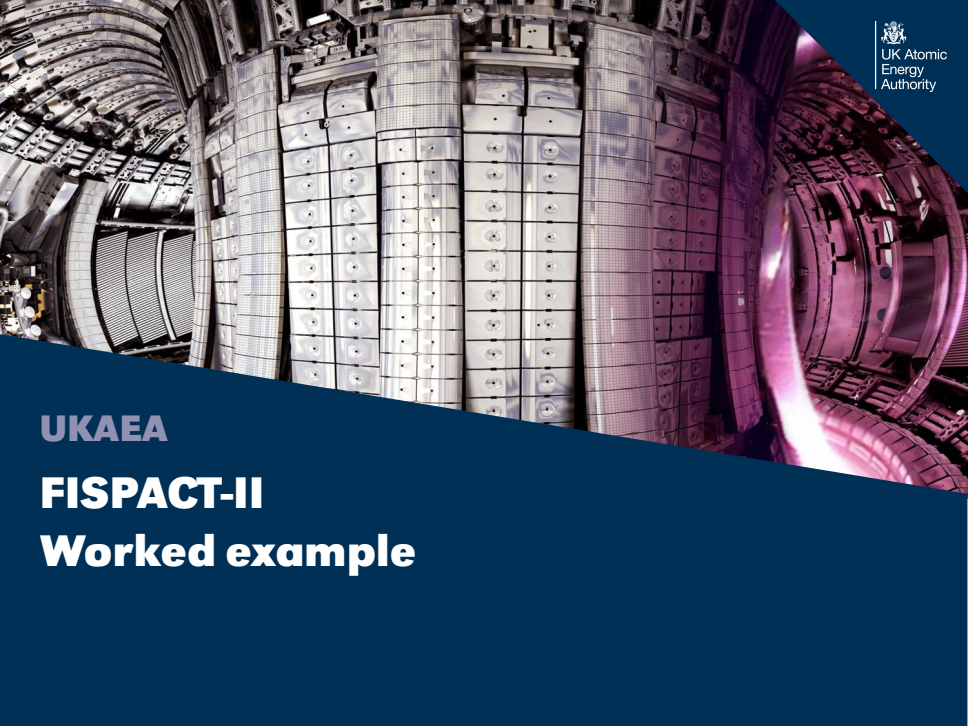




UKAEA

FISPACT-II

Worked example

Uncertainties and Pathways worked example: Irradiation of Iron

1. Investigate the production of Vanadium from irradiated iron.
2. Use Tendl17, GEFY6.1 and the fluxes from the FNS getting started example

Irradiation of Iron

```
CLOBBE
MONITOR 1
GETXS 1 709
GETDECAY 1
FISPACT
* Cobalt
MASS 1.0 1
FE 100.0
GRAPH 2 2 1 1 3
HALF
<< -----irradiation phase----- >>
FLUX 10.0E+10
ATOMS
TIME 10 DAYS
ATOMS
<< -----cooling phase----- >>
FLUX 0.
ZERO
TIME 1 HOURS ATOMS
END
* END
```

DOMINANT NUCLIDES

NUCLIDE	ACTIVITY (Bq)	PERCENT ACTIVITY	NUCLIDE	HEAT (kW)	PERCENT HEAT	NUCLIDE	DOSE RATE (Sv/hr)	PERCENT DOSE RATE
Total	7.6571E+10		Total	2.9192E-05		Total	3.5944E+01	
1 Mn 56	7.1822E+10	93.80E+00	Mn 56	2.9129E-05	99.79E+00	Mn 56	3.5853E+01	99.75E+00
2 Fe 55	3.2292E+09	42.17E-01	Mn 54	5.1226E-08	17.55E-02	Mn 54	8.2707E-02	23.01E-02
3 Cr 51	1.1338E+09	14.81E-01	Cr 51	6.6078E-09	22.64E-03	Cr 51	6.1371E-03	17.07E-03
4 Mn 54	3.8061E+08	49.71E-02	Fe 55	2.9308E-09	10.04E-03	Fe 53	1.0873E-03	30.25E-04
5 Fe 53	4.5920E+06	59.97E-04	Fe 53	1.6857E-09	57.75E-04	Fe 55	8.6845E-04	24.16E-04
6 Fe 59	6.5440E+05	85.46E-05	Fe 59	1.3708E-10	46.96E-05	Fe 59	2.2123E-04	61.55E-05
7 H 3	1.7145E+04	22.39E-06	Mn 52	5.1334E-12	17.59E-06	Mn 52	8.3281E-06	23.17E-06
8 Mn 52	9.0694E+03	11.84E-06	Mn 52m	4.9357E-13	16.91E-07	Mn 52m	5.3716E-07	14.94E-07
9 Mn 52m	8.6986E+02	1.136E-07	Cr 55	8.1711E-14	27.99E-08	Sc 48	4.3556E-10	12.12E-10
10 Cr 55	4.6332E+02	60.51E-08	H 3	1.5678E-14	53.71E-09	Cr 55	9.5540E-11	26.58E-11
11 Mn 53	1.9473E+02	25.43E-08	Sc 48	2.6725E-16	91.55E-11	Mn 53	4.3156E-11	12.01E-11
12 Sc 48	4.6734E-01	61.03E-11	Mn 53	1.6922E-16	57.97E-11	Mn 58	1.4668E-11	40.81E-12
13 Cr 56	1.0045E-01	13.12E-11	Mn 58	1.4337E-17	49.11E-12	Cr 49	9.4210E-12	26.21E-12
14 Cr 49	4.8130E-02	62.86E-12	Cr 49	1.2675E-17	43.42E-12	V 52	1.7030E-12	47.38E-13
15 V 49	4.5357E-02	59.24E-12	Cr 56	1.1240E-17	38.50E-12	Cr 56	2.2878E-13	63.65E-14
16 Mn 58	2.1860E-02	28.55E-12	V 52	1.6142E-18	55.30E-13	Co 58	1.5228E-13	42.37E-14
17 Co 58m	9.9510E-03	13.00E-12	Fe 52	1.2020E-19	41.18E-14	Fe 52	1.0342E-13	28.77E-14
18 V 52	4.0097E-03	52.37E-13	Co 58	1.0046E-19	34.41E-14	Sc 47	1.0896E-14	30.31E-15
19 Sc 47	1.3720E-03	17.92E-13	Sc 47	5.9560E-20	20.40E-14	V 49	6.5547E-15	18.24E-15
20 Fe 52	8.0296E-04	10.49E-13	Co 58m	3.8402E-20	13.15E-14	Co 58m	2.9137E-15	81.06E-16
Rest	6.2186E-04	81.21E-14	Rest	3.3533E-20	11.49E-14	Rest	1.0070E-15	28.02E-16

- We can see the production of an array of nuclei, Vanadium, Manganese etc
- Where did they come from?

Uncertainties and Pathways

UNCERTAINTY n (*path_floor loop_floor max_depth* n):

- $n = -1$ resets default values for a particular run and permits other values to be specified by the following parameters
- $n = 0$ no pathways or estimates of uncertainty are calculated
- $n = 1$ only estimates of uncertainty are output
- $n = 2$ both estimates of uncertainty and the pathway is output
- $n = 3$ only the pathway information is output
- cutoffs specified by *path_floor*, *loop_floor* and *max_depth*.
 - *path_floor*: pathways contributing more than the path floor fraction of the inventory of the final (target) nuclide are retained (default 0.005)
 - *loop_floor* All loops that increase the inventory contribution of the path they are on by a fraction greater than the loop floor are retained (default 0.01)
 - *max_depth* is the maximum number of links in a path from a source nuclide to a target nuclide (default 10)
- n following all the other parameters allows n to be input again so that after resetting cutoffs a calculation with the new values can be done.

UNCERTAINTY 2: Output

PATHWAY ANALYSIS FOR IRRADIATION PHASE

no of steps = 1
irradiation time = 8.64000E+05 secs
flux = 1.00000E+11 n/cm²/s

path floor = 5.00000E-01% of target inventory
loop floor = 1.00000E+00% of path inventory
max depth = 10 (maximum number of edges between source and target)

Pathways are given in order of decreasing dominance of target nuclides as ordered by the dominant nuclide tables above. Target nuclides at the end of the list that do not appear above are those that appear in the dominant lists at later cooling time. Pathways are retained if they contribute more than the path floor percentage of the number of target atoms given by the rate equation solution for the time interval. Loops are retained in a pathway if they contribute more than the loop floor percentage of the number of target atoms created along the pathway.

Pathways are analysed between the following source and target nuclides:

Source Nuclides

Fe 54 Fe 56 Fe 57 Fe 58

Target Nuclides

Mn 56 Mn 57 Fe 53 Mn 54 Mn 58 Cr 51 Fe 55 Mn 58m Cr 55 Fe 59
Mn 52 Mn 52m V 52 V 53 Cr 56 H 3 V 54 Sc 48 Ti 51 Mn 53
Cr 49

UNCERTAINTY 2: Output

Target nuclide Mn 56 100.000% of inventory given by 2 paths

path 1 99.465% Fe 56 ---(R)--- Mn 56 ---(S)---
100.00%(n,p)

path 2 0.545% Fe 57 ---(R)--- Mn 56 ---(S)---
87.43%(n,np)
12.57%(n,d)

Target nuclide Mn 57 99.573% of inventory given by 1 path

path 1 99.573% Fe 57 ---(R)--- Mn 57 ---(S)---
100.00%(n,p)

Target nuclide V 52 99.635% of inventory given by 4 paths

path 1 68.765% Fe 56 ---(R)--- Mn 55 ---(R)--- V 52 ---(S)---
91.37%(n,np) 100.00%(n,a)
8.63%(n,d)

path 2 24.809% Fe 56 ---(R)--- Cr 53 ---(R)--- V 52 ---(S)---
100.00%(n,a) 84.41%(n,np)
15.59%(n,d)

path 3 4.656% Fe 56 ---(R)--- Cr 52 ---(R)--- V 52 ---(S)---
100.00%(n,na) 100.00%(n,p)

path 4 1.405% Fe 56 ---(R)--- Fe 55 ---(B)--- Mn 55 ---(R)--- V 52 ---(S)---
100.00%(n,2n) 100.00%(b+) 100.00%(n,a)
0.00%(n,p)

- Two paths to produce Mn56
- Four paths to produce V52
- Edge types (r,R), (d,D) and (b,B) respectively denote reaction, decay and combined reaction and decay edges from short (lower case) and long lived (upper case) parents
- L and S denote short and long lived target nuclides.

Short-lived nuclides have half lives less than the time interval.

UNCERTAINTY 2: Output

UNCERTAINTY ESTIMATES (cross sections only)

Uncertainty estimates are based on pathway analysis for the irradiation phase

Total Activity is 1.00748E+11 +/- 2.00E+10 Bq. Error is 1.98E+01 % of the total.

Total Heat Production is 3.86884E-05 +/- 8.09E-06 kW. Error is 2.09E+01 % of the total.

Total Gamma Dose Rate is 4.71770E+01 +/- 9.95E+00 Sv/hr. Error is 2.11E+01 % of the total.

Total Gamma Heat Prod is 2.59475E-05 +/- 5.46E-06 kW. Error is 2.11E+01 % of the total.

Total Beta Heat Prod is 1.27409E-05 +/- 2.62E-06 kW. Error is 2.06E+01 % of the total.

0	Nuclide	Atoms	E(Atoms)	Activity	E(Activity)	Heat	E(Heat)	Dose Rate	E(Dose Rate)	% Error
Mn 56	1.25988E+15	2.67E+14	9.394E+10	1.99E+10	3.810E-05	8.09E-06	4.689E+01	9.95E+00	2.123E+01	
Mn 57	1.64432E+11	3.08E+10	1.335E+09	2.50E+08	2.574E-07	4.82E-08	2.726E-02	5.10E-03	1.871E+01	
Fe 53	4.44875E+11	8.89E+10	6.039E+08	1.21E+08	2.217E-07	4.43E-08	1.430E-01	2.86E-02	1.999E+01	
Mn 54	1.48097E+16	1.55E+15	3.806E+08	4.00E+07	5.123E-08	5.38E-09	8.271E-02	8.68E-03	1.050E+01	
Mn 58	3.41685E+09	1.46E+09	3.632E+07	1.56E+07	2.382E-08	1.02E-08	2.437E-02	1.04E-02	4.283E+01	

- The uncertainties of the total values and the dominant nuclides

LOOKAHEAD

For a simulation with a long cooling time the dominant nuclides at late times may not be significant at the end of the irradiation phase, leading to poor estimates for the uncertainties. One remedy for this is to use the LOOKAHEAD.

LOOKAHEAD

For example including LOOKAHEAD in our simulation shows that Sc 47 is present at the end of our simulation (small quantity):

PATHRESET

LOOKAHEAD can lead to an excessively large numbers of target nuclides in the pathways calculations, hence a long compute time. PATHRESET forces the pathways calculation to be repeated at the cooling step immediately before its occurrence

- The PATH keyword allows a particular pathway consisting of link reactions and decays to be shown.

For example:

```
MASS 1.0 1
FE 100.0
GRAPH 2 2 1 1 3
HALF
PATH 2
Fe57 X Cr54 X V54
<< -----irradiation phase----- >>
FLUX 10.0E+10
ATOMS
TIME 10 DAYS
ATOMS
<< -----cooling phase----- >>
FLUX 0.
ZERO
TIME 1 HOURS ATOMS
```

```
PATHS ANALYSIS FOR IRRADIATION PHASE
=====

no of steps      =      1
irradiation time = 8.64000E+05 secs
flux            = 1.00000E+11 n/cm**2/s

Target nuclide V 54      78.860% of inventory given by 1 path
=====

path 1 78.860% Fe 57 ----(R)---- Cr 54 ----(R)---- V 54 ----(S)----
100.00%(n,a)      100.00%(n,p)
1 * * * TIME INTERVAL 3 * * * COOLING TIME IS 3.6000E+03 SECS OR 1.0000E+00 HOURS * * * ELAPSED TIME IS
1.000 h * * * FLUX AMP IS 0.0000E+00 /cm^2/s * * *
```

PATH is only necessary if a special investigation of a certain pathway is needed. Otherwise use UNCERTAINTY.

ROUTES *par dau path_floor loop_floor max_depth*

- searches for all pathways from the parent nuclide *par* to the daughter nuclide *dau*
- cutoffs specified by *path_floor*, *loop_floor* and *max_depth*.

```
MASS 1.0 1
FE 100.0
GRAPH 2 2 1 1 3
HALF
ROUTES FE57 V54 1.0e-4 1.0e-2 25
<< -----irradiation phase----- >>
FLUX 10.0E+10
ATOMS
TIME 10 DAYS
ATOMS
<< -----cooling phase----- >>
FLUX 0.
ZERO
TIME 1 HOURS ATOMS
END
```

```
flux = 1.00000E+11 n/cm..2/s
path floor = 1.00000E-02% of target inventory
loop floor = 1.00000E+00% of path inventory
max depth = 25 (maximum number of edges between source and target)
```

Source Nuclides

Fe 57

Target Nuclides

V 54

Target nuclide V 54 78.835% of inventory given by 1 path

path 1 78.835% Fe 57 --- (R) --- Cr 54 --- (R) --- V 54 --- (S) ---

100.00%(n,a) 100.00%(n,p)

t . . . TIME INTERVAL 3 . . . COOLING TIME IS 3.6000E+03 SECS OR 1.0000E+00 HOURS . . . ELAPSED TIME IS 1.000 h

. . . FLUX AMP IS 0.0000E+00 /cm^2/s . . .

NUCLIDE	ATOMS	GRAMS	Bq	b-Energy MW	a-Energy MW	g-Energy MW	DOSE RATE Sv/hr	HALF LIFE seconds
---------	-------	-------	----	----------------	----------------	----------------	--------------------	----------------------

H 1	# 1.81324E+17	3.035E-07	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.000E+00	Stable
-----	---------------	-----------	-----------	-----------	----------	-----------	-----------	--------

We learned about:

- **UNCERTAINTY**
 - Show paths and uncertainties
- **LOOKAHEAD**
 - Produces uncertainties relevant for long cooling time
- **PATHRESET**
 - Resets pathway calculations reducing compute overhead
- **PATH**
 - Identifies specified paths